

Question One	12 marks	(Start a new booklet)	Marks
a)	Factorise $45x^2 - 80y^2$		2
b)	Solve for x : $ 2x+1 =3$		2
c)	If α and β are the roots of $3x^2 - 5x + 1 = 0$, find the value $\frac{1}{\alpha} + \frac{1}{\beta}$		2
d)	Express $x - \frac{1}{x}$ as an exact value in its simplest form if $x = 1 - \sqrt{5}$		2
e)	Simplify $(x+2)^2 - (x+3)(x-3)$		2
f)	Evaluate $\lim_{x \rightarrow 2} \frac{x^2 - 6x + 8}{x^2 - 4}$		2

Question Two	12 Marks	(Start a new booklet)	Marks
a)	The triangle ABC has vertices $A(-2, 3)$, $B(4, 1)$ and $C(2, -5)$		
i)	Find the co-ordinates of the midpoint of BC		1
ii)	Find the equation of the line passing through C which is parallel to AB		2
iii)	Prove that ABC is a right-angled triangle.		2
b)	The line L has equation $y + 2x = 12$ and the curve C has equation $y = x^2 - 4x + 9$		
i)	By completing the square, state the co-ordinates of the minimum point of the curve C .		2
ii)	Find the co-ordinates of the point of intersection of L and C .		3
c)	The line AB has equation $3x + 2y = -21$. Find the perpendicular distance from the origin to the line AB , in surd form.		2

Question Three	12 Marks	(Start a new booklet)	Marks
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a) Differentiate the following functions:

i) $y = (7 - 3x)^6$			2
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ii) $y = x \tan x$			2
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iii) $y = \frac{x}{\sin 2x}$			2
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b) Evaluate the following integrals

i) $\int_1^4 \sqrt{x} \, dx$			2
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ii) $\int_0^2 e^{3x} \, dx$			2
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c) Find $\int \frac{6x}{x^2 + 3} \, dx$			2
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Question Four	12 Marks	(Start a new booklet)	Marks
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a) In an arithmetic sequence, the sixth term is 13 and the tenth term is 1:

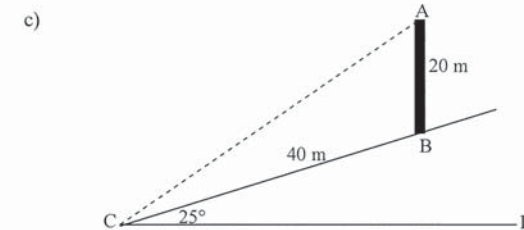
i) Find the first term and the common difference.			2
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ii) Find the sum of the first twenty terms.			2
---	--	--	---

b) A container holds 50 litres of oil. A pump withdraws 10 litres on the first stroke and 7.5 litres on the second stroke. On each future stroke, the pump withdraws $\frac{3}{4}$ of the amount of the previous stroke:

i) show that the container will never be emptied			1
--	--	--	---

ii) find how much oil will finally remain in the container			2
--	--	--	---



A 20 metre high vertical mast AB is placed 40 metres from the base C of a slope inclined at 25° to the horizontal. A wire support AC is used to keep the mast in position:

i) Explain why angle $ABC = 115^\circ$			1
--	--	--	---

ii) Show that the length of the wire AC is 51.7 m (to 1 decimal place)			2
--	--	--	---

iii) Hence find the angle which the wire AC makes with the slope CB . Answer to the nearest degree.			2
---	--	--	---

Question Five **12 Marks** **(Start a new booklet)** **Marks**

- a) i) Copy and complete this table for $f(x) = x e^x$, giving values to 2 decimal places.

x	0	1	2
$f(x)$			

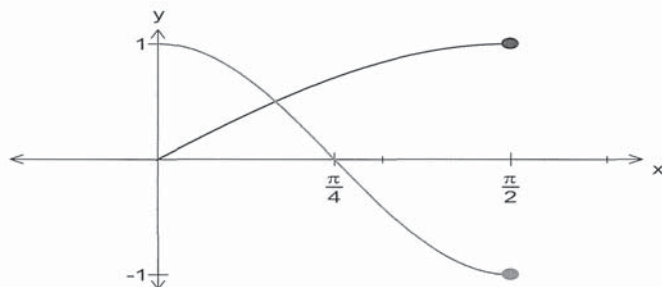
1

- ii) Use Simpson's rule to estimate the value of $\int_0^2 x e^x dx$

2

- b) The diagram shows the graphs of $y = \sin x$ and $y = \cos 2x$ for $0 \leq x \leq \frac{\pi}{2}$.

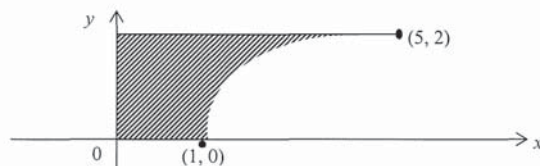
The graphs intersect at $A\left(\frac{\pi}{6}, \frac{1}{2}\right)$



Find the area of the shaded region.

4

- c) The diagram shows the graph of $y = \sqrt{x-1}$ between $(1, 0)$ and $(5, 2)$.



The shaded region is rotated about the y -axis. Find the volume of the solid formed.

5

Question Six **12 Marks** **(Start a new booklet)** **Marks**

- a) Solve for x : $3^x \times 9^{x+1} = \frac{1}{3}$

2

- b) In the diagram $ABCD$ is a straight line, and E lies on CF . $BF = EF$, $\angle BFE = 44^\circ$, $\angle DCE = 146^\circ$ and $\angle CBE = x^\circ$

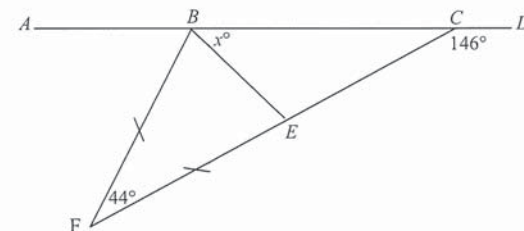


Diagram
not
to
scale

- i) Find the value of x giving reasons.

3

- ii) State why $BE = EC$.

1

- c)

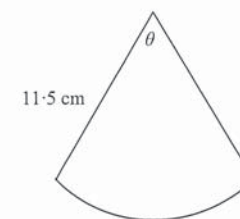


Diagram
not
to
scale

The radius of a sector of a circle is 11.5 cm, and its perimeter is 36.8 cm:

- i) Find the size of the angle θ to the nearest degree

2

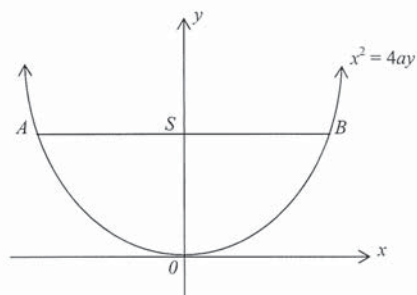
- ii) Find the area of the sector.

1

Question Six continued

Marks

d)

Diagram
not
to
scale

The diagram shows the graph of the parabola $x^2 = 4ay$, with focus S , and AB is the latus rectum (that is, the focal chord parallel to the x -axis of the parabola).

Prove that the length of the latus rectum is $4a$ units.

3

Question Seven on the next page

Question Seven

12 Marks

(Start a new booklet)

Marks

- a) A particle moves in a straight line. At time t seconds, its distance is x metres from a fixed point O and its velocity is given by the equation

$$v = 4t - 3t^2$$

Initially the particle is at $x = 3$

- i) Find the position of the particle when $t = 2$

3

- ii) Find the velocity of the particle when the acceleration is zero

2

- iii) Find the acceleration of the particle when the particle becomes instantaneously at rest during the motion.

2

- b) The rate of flow of water into a large container is given by:

$$\frac{dv}{dt} = \frac{30}{t+1}$$

where v is in litres and t is in minutes.

Initially, there are 40 litres of water in the container.

- i) Find the volume of water in the container after 4 minutes.

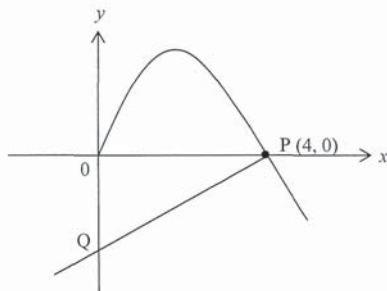
3

- ii) How long does it take for the container to hold 160 litres.

2

Question Eight **12 marks** **(Start a new booklet)** **Marks**

- a) A curve $y = 4x^{1/2} - x^{3/2}$ defined for $x \geq 0$, crosses the x -axis at the point $P(4, 0)$ the normal to the curve at P meets the y -axis at the point Q , as shown in the diagram below:



- i) Show that the gradient of the curve at $P(4, 0)$ is -2 1
- ii) Find the area of triangle OPQ 3
- iii) Find the total area of the region bounded by the curve and the lines PQ and QO . 2
- b) A particle moves in a straight line with a constant acceleration of 2m/sec^2 towards the right. It is initially at rest, 2 metres to the right of the origin:
- i) Find an expression for the velocity after t seconds 1
- ii) Find the velocity after 3 seconds 1
- iii) When is the velocity 10m/sec ? 1
- iv) Find an expression for the displacement after t seconds. 1
- v) When is the particle 6 metres to the right of the origin. 1
- vi) How far does the particle move in the third second. 1

Question Nine **12 Marks** **(Start a new booklet)** **Marks**

- a) On the birth of Fernando, his parents decide to set up an investment fund for him. They agree to deposit \$500 on the 18th September 2008 (the date of birth) and \$500 on every subsequent birthday up to and including his 21st birthday. Interest is being paid at the rate 6% p.a. to be compounded:
- i) How much will the first \$500 deposit mature to after 21 years? 2
- ii) How much will Fernando receive on his 21st birthday? 2
- b) Find the volume of the solid of revolution formed by rotating the curve $y = e^x + e^{-x}$ about the x -axis between $x = -1$ and $x = 1$, leaving your answer in exact form: 4
- c) i) Find the x co-ordinate of the point on the graph of $y = 5 - 14x - 2x^2$ where the tangent is parallel to the line $2x + y - 3 = 0$ 2
- ii) Hence find the equation of this tangent from part (i) 2

Question Ten 12 Marks (Start a new booklet)

Marks

Consider the function $f(x) = \frac{e^x}{x}$ i) What is the domain of $f(x)$?

1

ii) The first derivative of $f(x)$ is $f'(x) = \frac{xe^x - e^x}{x^2}$

Show that the second derivative can be written as:

$$f''(x) = \frac{e^x [(x-1)^2 + 1]}{x^3}$$

2

iii) Find the co-ordinates of the stationary point and determine its nature

3

iv) Show that there are no points of inflexion

1

v) For what values of x is the curve concave up and concave down?

2

vi) Discuss briefly what happens to $f(x)$ when $x \rightarrow \infty$

1

vii) Sketch the graph of $y = f(x)$

2

END OF TRIAL EXAMINATION

YEAR 12 2unit Advanced Trial Solⁿ 2010

Question 1

$$a) \frac{45x^2 - 80y^2}{5(9x^2 - 16y^2)}$$

$$\Rightarrow [(3x+4y)(3x-4y)]$$

$$\frac{\alpha + \beta}{\alpha \beta} = \frac{\frac{5}{3}}{\frac{1}{3}}$$

$$\frac{5}{3} \times \frac{3}{1} = 5$$

$$b) |2x+1| = 3$$

$$\pm (2x+1) = 3$$

$$2x+1 = 3$$

$$2x = 2$$

$$x = 1 \text{ check}$$

or

$$-2x-1 = 3$$

$$-2x = 4$$

$$2x = -4$$

$$x = -2 \text{ check}$$

$$c) 3x^2 - 5x + 1 = 0$$

$$\frac{1}{\alpha} + \frac{1}{\beta}$$

$$\Rightarrow \frac{\alpha + \beta}{\alpha \beta}$$

$$\alpha + \beta = \frac{5}{3} \quad \alpha \beta = \frac{1}{3}$$

$$d) x - \frac{1}{x} \quad x = 1 - \sqrt{5}$$

$$\Rightarrow 1 - \sqrt{5} - \frac{1}{1 - \sqrt{5}} \times \frac{1 + \sqrt{5}}{1 + \sqrt{5}}$$

$$\Rightarrow 1 - \sqrt{5} - \frac{(1 + \sqrt{5})}{-4}$$

$$\Rightarrow \frac{5 - 3\sqrt{5}}{4}$$

$$e) (x+2)^4 - (x+3)(x-3)$$

$$\Rightarrow (x^2 + 4x + 4) - (x^2 - 9)$$

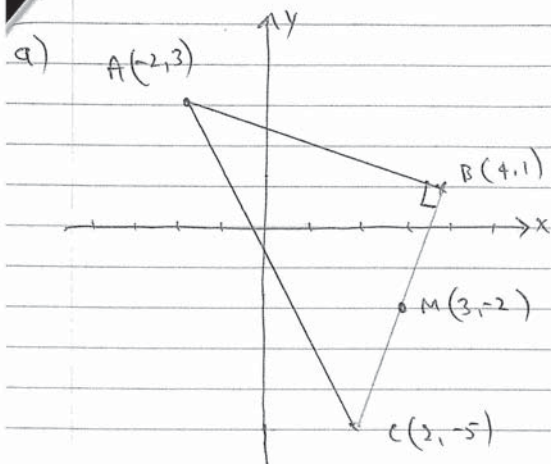
$$\Rightarrow 4x + 13$$

$$f) \lim_{x \rightarrow 2} \frac{x^2 - 6x + 8}{x^2 - 4}$$

$$\lim_{x \rightarrow 2} \Rightarrow \frac{(x-4)(x-2)}{(x+2)(x-2)}$$

$$\lim_{x \rightarrow 2} \frac{x-4}{x+2} = -\frac{1}{2}$$

Question 2



i) Midpoint BC

$$M\left[\frac{4+2}{2}, \frac{1+(-5)}{2}\right]$$

$$M(3, -2)$$

ii) A(-2, 3) B(4, 1)

$$m(AB) = \frac{3-1}{-2-4}$$

$$= \frac{2}{-6}$$

$$m(AB) = -\frac{1}{3}$$

$$m = -\frac{1}{3} \quad C(2, -5)$$

$$y+5 = -\frac{1}{3}(x-2) \quad \times 3$$

$$3y+15 = -(x-2)$$

$$3y+15 = -x+2$$

$$\boxed{x+3y+13=0}$$

iii) $m_1(AB) = -\frac{1}{3}$

$m_2(BC) \quad B(4, 1) \quad C(2, -5)$

$$m_2(BC) = \frac{1-(-5)}{4-2}$$

$$m_2 = \frac{6}{2} = 3$$

$$m_1 \times m_2 = -1$$

$$-\frac{1}{3} \times 3 = -1$$

$$\therefore AB \perp BC$$

b) $x^2 - 4x + 9 = 0$

i) $x^2 - 4x = -9$

$$x^2 - 4x + (-2)^2 = -9 + (-2)^2$$

$$(x-2)^2 = -5$$

$$(x-2)^2 + 5 = y$$

Vertex (2, 5) Min TP.

ii) $y = x^2 - 4x + 9 \quad y = 12 - 2x$

$$12 - 2x = x^2 - 4x + 9$$

$$0 = x^2 - 2x - 3$$

$$(x+1)(x-3)$$

$$x = -1 \quad x = 3$$

$$y = 14 \quad y = 6$$

$$(3, 6) \quad (-1, 14)$$

c) $3x + 2y = -21 \quad \text{Origin } (0, 0)$

$$\frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}} = \frac{21}{\sqrt{13}} \times \frac{\sqrt{13}}{\sqrt{13}} = \frac{21\sqrt{13}}{13}$$

Question 3

i) $y = (7-3x)^6$

$$y' = 6(7-3x)^5 \times -3$$

$$y' = -18(7-3x)^5$$

ii) $y = x \tan x$

$$y = u \cdot v$$

$$u = x \quad v = \tan x$$

$$u' = 1 \quad v' = \sec^2 x$$

$$y' = x \cdot \sec^2 x + \tan x \cdot 1$$

$$y' = x \sec^2 x + \tan x$$

$$y = \frac{x}{\sin 2x}$$

$$y' = \frac{\sin 2x - 2x \cos 2x}{\sin^2 2x}$$

$$y = \frac{u}{v}$$

$$u = x \quad v = \sin 2x$$

$$u' = 1 \quad v' = 2 \cos 2x$$

$$y' = \frac{\sin 2x \cdot 1 - x \cdot 2 \cos 2x}{(\sin 2x)^2}$$

$$b) i) \int_1^4 \sqrt{x} \, dx \rightarrow \left(\frac{2}{3} \sqrt{x^3} \right) - \frac{2}{3} \sqrt{1^3}$$

$$\int_1^4 x^{1/2} \, dx \Rightarrow \left(\frac{2}{3} x^{3/2} \right) - \frac{2}{3}$$

$$\left[\frac{2}{3} x^{3/2} \right]_1^4 = \frac{16}{3} - \frac{2}{3} = \frac{14}{3}$$

$$ii) \int_0^2 e^{3x} \, dx \quad y = e^{3x} \\ y' = 3e^{3x}$$

$$\Rightarrow \left[\frac{1}{3} e^{3x} \right]_0^2$$

$$\frac{1}{3} (e^6 - e^0)$$

$$\frac{1}{3} (e^6 - 1)$$

$$\int 3e^{3x} \, dx \\ = e^{3x}$$

$$iii) \int \frac{6x}{x^2+3} \, dx \Rightarrow 3 \log(x^2+3) + C$$

Question 4

$$a) T_6 = 13 \quad T_{10} = 1$$

$$i) a + 5d = 13 \quad a + 9d = 1$$

Solve

$$a + 9d = 1$$

$$a + 5d = 13$$

$$4d = -12$$

$$d = -3$$

Sub to find a

$$a + 5d = 13$$

$$a - 15 = 13$$

$$a = 28$$

$$d = -3$$

ii) Sum of first 20 terms

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

$$a = 28$$

$$n = 20$$

$$S_{20} = 10 [56 + (19) \times -3]$$

$$d = -3$$

$$= 10 [56 - 57]$$

$$S_{20} = -10$$

$$b) S_n = 10 + 7.5 + \dots$$

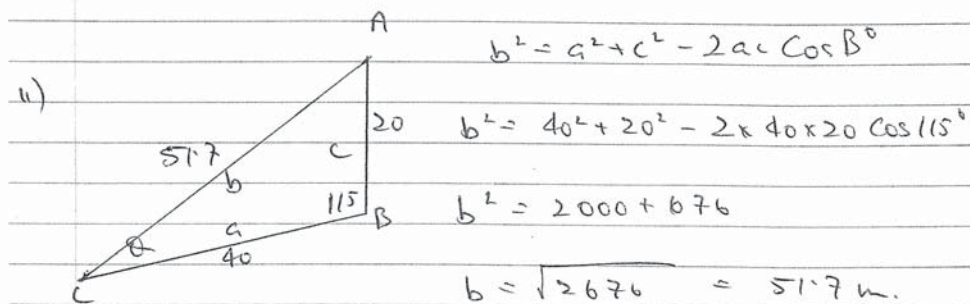
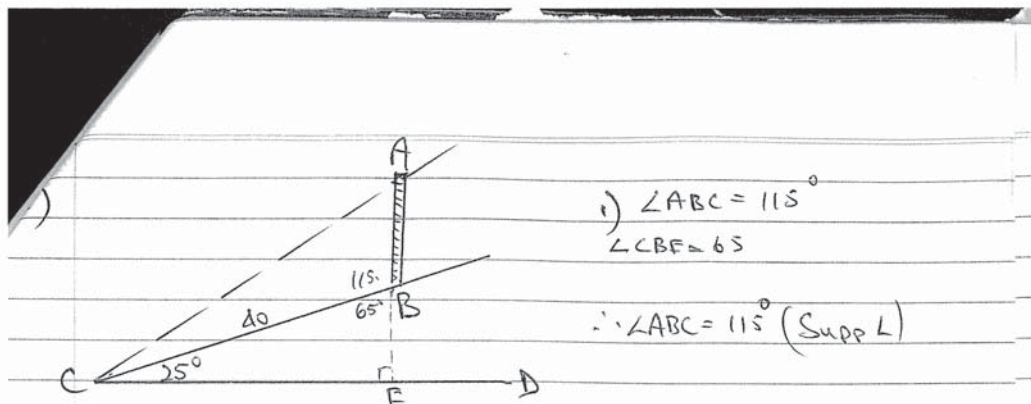
$$S_n = \frac{10}{\frac{1}{4}} = 40$$

$$S_n = \frac{a}{1-r}$$

$$= \frac{10}{1 - \frac{3}{4}}$$

limiting sum is 40

$\therefore$ 10 litres remain



iii) $\frac{b}{\sin B} = \frac{c}{\sin C^\circ}$

$\frac{51.7}{\sin 115} = \frac{20}{\sin \theta}$

$51.7 \sin \theta = 20 \sin 115^\circ$

$\sin \theta = \frac{20 \sin 115^\circ}{51.7}$

$\sin \theta = 0.365739007$

$\theta = 21^\circ$

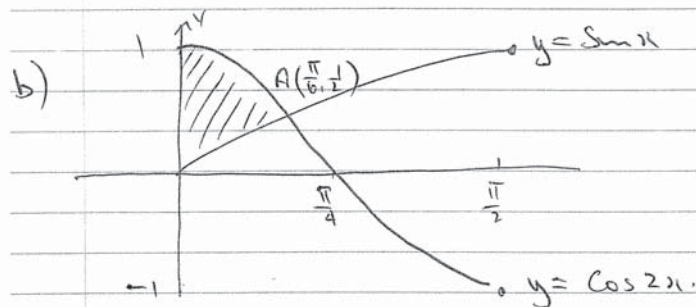
Question 5

a)

x	0	1	2
f(x)	0	2.72	14.78

$f(x) = x e^x$

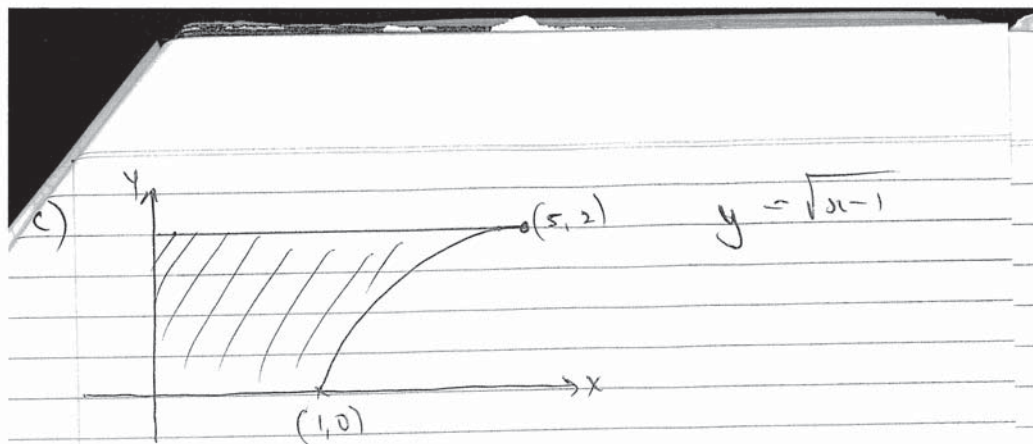
$\int_0^2 x e^x dx \doteq \frac{1}{3} [0 + 4 \times 2.72 + 14.78]$
 $\doteq 8.55$ (2 dec pl)



$A = \int_0^{\frac{\pi}{6}} (\cos 2x - \sin x) dx$
 $= \left[\frac{1}{2} \sin 2x + \cos x \right]_0^{\frac{\pi}{6}}$

$\Rightarrow \left(\frac{1}{2} \times \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} \right) - (0 + 1)$

$\Rightarrow \frac{3\sqrt{3} - 1}{4} \text{ unit}^2$



$$y = \sqrt{x-1}$$

$$y^2 = x-1$$

$$x = y^2 + 1 \quad \text{Make } x\text{-subject \{ Rotated about } y \}}$$

$$V = \pi \int_0^2 (y^2 + 1)^2 dy$$

$$V = \pi \int_0^2 (y^4 + 2y^2 + 1) dy$$

$$= \pi \left[\frac{1}{5} y^5 + \frac{2}{3} y^3 + y \right]_0^2$$

$$= \pi \left[\frac{32}{5} + \frac{16}{3} + 2 \right]$$

$$V = \frac{206\pi}{15} \text{ unit}^3$$

Question 6

a) $3^x \times 9^{x+1} = \frac{1}{3}$

$$3^x \times (3^2)^{x+1} = 3^{-1}$$

$$3^x \times 3^{2x+2} = 3^{-1}$$

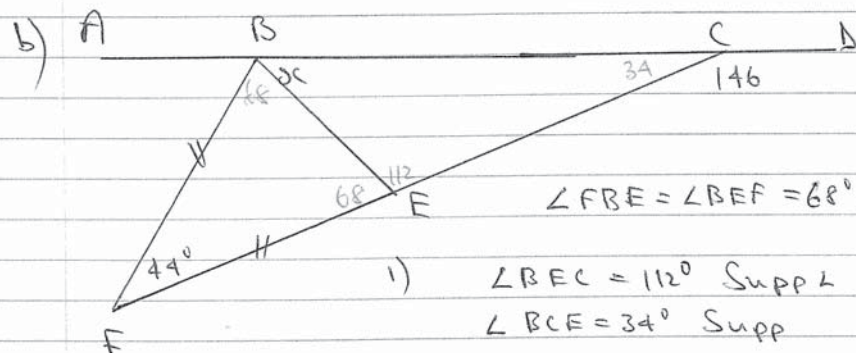
$$3^{3x+2} = 3^{-1}$$

equating indices

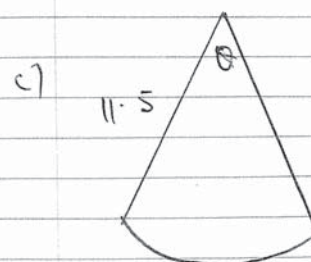
$$3x+2 = -1$$

$$3x = -3$$

$$\boxed{x = -1}$$



ii) Base angle $= 34^\circ \therefore$ isos.



i) $P = r\theta + 2r$

$$36.8 = 11.5\theta + 23$$

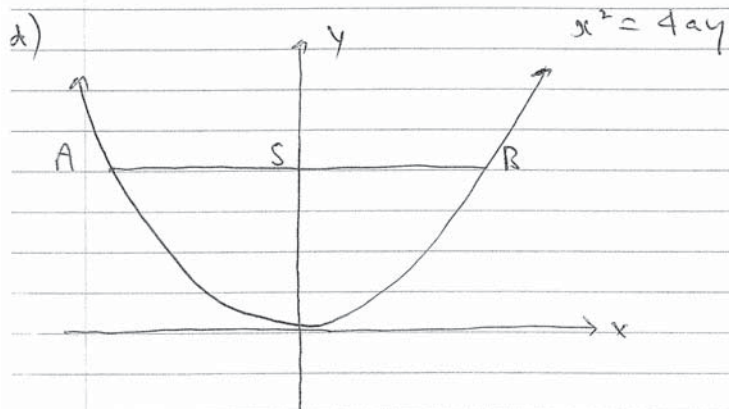
$$11.5\theta = 13.8$$

$$\theta = \frac{13.8}{11.5}$$

$$\theta = 1.2 \text{ Radians}$$

$1.2 \times \frac{180}{\pi} = 69^\circ$

$$\begin{aligned}\text{Area of Sector} &= \frac{1}{2} r^2 \theta \\ &= \frac{1}{2} \times 11.5^2 \times 1.2 \\ A &= 79.35 \text{ cm}^2\end{aligned}$$



Prove $AB = 4a$ unit

Focus $S(0, a)$ $\therefore AB \Rightarrow y = a$

Substitute $y = a$ into $x^2 = 4ay$

$$\begin{aligned}\Rightarrow x^2 &= 4a \cdot a \\ x^2 &= 4a^2\end{aligned}$$

$$x = \sqrt{4a^2}$$

$$x = \pm 2a$$

$\Rightarrow 4a$ units

Question 7

a) $V = 4t - 3t^2$

i) initially particle is at $x=3$
ie at $t=0$ $x=3$

$$V = 4t - 3t^2$$

$$x = \int 4t - 3t^2 dt$$

$$x = \frac{4t^2}{2} - \frac{3t^3}{3} + c$$

$$x = 2t^2 - t^3 + c$$

$$\text{at } t=0 \quad x=3$$

$$\therefore 3 = 0 + c \quad \therefore c = 3$$

$$x = 2t^2 - t^3 + 3$$

$$\text{at } t=2$$

$$x = 8 - 8 + 3 \quad \boxed{x = 3 \text{ m}}$$

ii) $V = 4t - 3t^2$

$$\Rightarrow 6t = 4$$

$$t = \frac{4}{6}$$

$$a = \frac{dV}{dt} = 4 - 6t$$

$$\text{at } a=0$$

$$t = \frac{2}{3} \text{ sec.}$$

$$0 = 4 - 6t$$

$$V = 4t - 3t^2 \quad \text{at } t = \frac{2}{3}$$

$$V = 4\left(\frac{2}{3}\right) - 3\left(\frac{2}{3}\right)^2$$

$$= \frac{8}{3} - 3\left(\frac{4}{9}\right)$$

$$= \frac{8}{3} - \frac{12}{9}$$

$$\frac{24}{9} - \frac{12}{9}$$

$$= \frac{12}{9}$$

$$= \frac{4}{3} \text{ or } 1\frac{1}{3} \text{ m/sec.}$$

$$1) \frac{dV}{dt} = \frac{30}{t+1}$$

$$V = \int \frac{30}{t+1} dt$$

$$V = 30 \ln(t+1) + c$$

$$\text{at } t=0 \quad V=40$$

$$40 = 30 \ln 1 + c$$

$$\therefore c = 40$$

$$V = 30 \ln(t+1) + 40$$

find Volume at $t=4$.

$$V = 30 \ln(5) + 40$$

$$V = 88.3 \text{ litres.}$$

$$b) 1) \int \frac{30}{t+1} dt$$

$$V = 30 \ln(t+1) + c$$

$$\text{at } t=0 \quad V=40$$

$$40 = 30 \ln 1 + c$$

$$\therefore c = 40$$

$$V = 30 \ln(t+1) + 40$$

$$120 = 30 \log_e(t+1)$$

$$4 = \log_e(t+1)$$

$$\log_e y = x \rightarrow y = e^x$$

$$\log_e(t+1) = 4$$

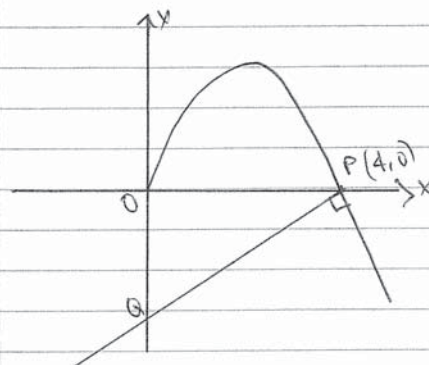
$$t+1 = e^4$$

$$t = e^4 - 1$$

$$11) 160 = 30 \ln(t+1) + 40$$

Question 8

a)



$$y = 4x^{1/2} - x^{3/2}$$

$$\frac{dy}{dx} = 2x^{-1/2} - \frac{3}{2}x^{1/2}$$

$$\frac{dy}{dx} = \frac{2}{x^{1/2}} - \frac{3}{2}x^{1/2}$$

$$\text{at } x=4$$

$$m = \frac{2}{2} - \frac{3 \times 2}{2}$$

$$m = 1 - 3 \quad \boxed{m = -2}$$

11) Area of triangle OPQ

$$m_{PQ} = \frac{1}{2} \quad P(4,0)$$

$$y-0 = \frac{1}{2}(x-4)$$

$$2y = x - 4$$

$$2y = x - 4$$

$$-x + 2y = -4$$

$$x - 2y = 4$$

$$\text{at } x=0 \quad \text{cut } y = -2$$

$$\therefore -2y = 4$$

$$y = -2$$

$$Q(0, -2)$$

$$A = \frac{1}{2}bh$$

$$= \frac{1}{2} \times 2 \times 4$$

$$= 4 \text{ m}^2$$

iii) Find area bounded by curve and the lines PQ and QO

Region bounded by curve and x-axis.

$$A = \int_0^4 4x^{1/2} - x^{3/2} dx$$

$$= \frac{2}{3} \cdot 4x^{3/2} - \frac{2}{5}x^{5/2}$$

$$= \left[\frac{8}{3}x^{3/2} - \frac{2}{5}x^{5/2} \right]_0^4$$

$$= \frac{8 \times 8}{3} - \frac{64}{5}$$

$$= \frac{128}{15} \text{ units}$$

$$\text{Area Region} = \frac{128}{15} + 4$$

$$= 12 \frac{8}{15} \text{ unit}^2.$$

b) $a = 2 \text{ m/sec}^2$, it is initially at rest 2m to right of origin

$$a = 2$$

$$v = \int 2 dt$$

$$v = 2t + c$$

at $t=0$ $v=0$

$$v = 2t$$

$$v) V = 2t \text{ at } t=3$$

$$V = 6 \text{ m/sec}$$

iii) When is velocity 10 m/sec

$$V = 2t$$

$$10 = 2t$$

$$t = 5 \text{ secs.}$$

iv) displacement

$$V = 2t$$

$$x = \int 2t dt$$

$$x = \frac{2t^2}{2}$$

$$x = t^2 + c$$

$$\text{at } t=0 \quad x=2$$

$$\therefore c = 2$$

$$x = t^2 + 2$$

$$x = t^2 + 2$$

$$6 = t^2 + 2$$

$$t^2 = 4 \quad \therefore t = 2 \text{ sec.}$$

vi) How far in 3rd second.

$$x = t^2 + 2 \quad t=2 \quad x=6$$

$$x = t^2 + 2 \quad t=3 \quad x=11$$

$$\therefore 5 \text{ metres}$$

Question 9

a) $A = 500(1.06)^{21}$

$= \$1699.78$

ii) $\frac{500 \times 1.06(1.06^{21}-1)}{1.06-1}$

$= \$21,196.15 \rightarrow \500

$= \$26,696.15$

b) $y = e^x + e^{-x}$

$V = \pi \int_{-1}^1 (e^x - e^{-x})^2 dx$

$= \pi \int_{-1}^1 e^{2x} + 2 + e^{-2x} dx$

$= \pi \left[\frac{1}{2} e^{2x} + 2x - \frac{1}{2} e^{-2x} \right]_{-1}^1$

$= \pi \left[\frac{1}{2} e^2 + 2 - \frac{1}{2} e^{-2} \right] - \left[\frac{1}{2} e^{-2} - 2 - \frac{1}{2} e^2 \right]$

$= \pi [e^2 + 4 + e^{-2}]$

$= \pi (11.53)$

$= 36.2 \text{ m}^3$

c) $y = 5 - 14x - 2x^2$

$y' = -14 - 4x$

$-2 = -14 - 4x$

$4x = -12$

$x = -3$

Sub to find y

$y = 29$

$(-3, 29)$

$y - 29 = -2(x + 3)$

$y - 29 = -2x - 6$

$2x + y - 23 = 0$

Question 10
1) Domain $x \neq 0$ all real

ii) $f'(x) = \frac{xe^x - e^x}{x^2}$

$f''(x) = \frac{x^2 [e^x + xe^x - e^x] - [xe^x - e^x] 2x}{x^4}$

$= \frac{x^3 e^x - 2x^2 e^x + 2x e^x}{x^4}$

$= \frac{xe^x (x^2 - 2x + 2)}{x^4}$

$f''(x) = \frac{e^x [(x-1)^2 + 1]}{x^3}$

iii) Stationary Point $f'(x) = 0$

$\therefore xe^x - e^x = 0$

$e^x (x-1) = 0$

$\therefore \text{at } x=1$

Sub to find y

$y = \frac{e^x}{x} \text{ at } x=1$

$y = e$

Nature $f''(x)$

$f''(1) = \frac{e^1 \cdot 1}{1} > 0$

$\therefore \text{Min TP } (1, e)$

iv) No pts of inflexion $f''(x) = 0$

$$\text{ie } (x-1)^2 + 1 > 0 \text{ for all values of } x$$

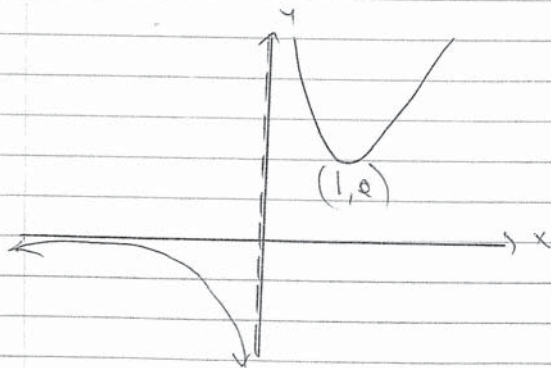
ie $f''(x) \neq 0 \therefore$ No pt inflexion

v) Concave up $f''(x) > 0$ occurs for $x > 0$

Concave down $f''(x) < 0$ occurs for $x < 0$

vi) $\lim_{x \rightarrow \infty} \frac{e^x}{x} \Rightarrow 0$ tends to ~~∞~~ for large values of x .

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END OF SOLUTIONS